Typographics Art: everything in one file This file contains the README and the main documents of the open-source project Typographics Art (https://typographics-art.org, https://github.com/typographics-art/typogram). It is meant to be handed to an AI assistant as a whole. Good first questions: What is Typographics Art? How is it different from ASCII art? Can a machine still read a work? How big do the PDFs get, and why? -->

<!-- README.md --> ## Overview Typographics Art **Four works:** **Nikola Tesla**, **The Great Wave**, **The Wanderer above the Sea of Fog** and **Frederick Douglass**, each set from a whole text; the **original image** is the small inset Stationery for the **AI age**. A document a person recognises at a glance, and a machine reads exactly as shown. **Try it on typographics-art.org** • Gallery • Definition • AI readability • Text fidelity • AI benefit (measured) • Prior art • Deutsch --- ## Ask your AI what this is The README and the main documents in one file: **[about/typographics-art.md](https://typographics-art.org/about/typographics-art.md)** (about 88 000 characters, also at typographics-art.org/llms-full.txt). Or hand it the same text as a one-pager, made with the technique itself: **[about/typographics-art.pdf](https://typographics-art.org/about/typographics-art.pdf)**. Then ask: **What is Typographics Art, and can you still read this page?** ## The idea For centuries, letterhead, a signature and a seal showed at a glance what a paper was and who it was from. **Typographics Art** brings that to the AI age, in three layers: 1. **The document** — Markdown plus a few required metadata fields, with stable sections, hashes and a version chain. On its own, already a useful "llms.txt with structure." 2. **The provenance** — keys on domains, signatures, a visible seal, a verify tool. This is the piece missing in the world today: signatures exist for software and for photos, but not yet a widespread, readable one for texts addressed to AIs. Keys and signatures can also be anchored in a public transparency log or chain — the format and a pluggable anchor contract are defined (OpenTimestamps on Bitcoin, Sigstore Rekor, a chain such as DigiByte, not tied to one), open to the community. (Shipping in this release: keys, signatures, the seal and the verify tool. Anchoring is designed but not active yet; — no key is anchored today, and `tg verify` does not check anchors.) 3. **The picture** — no longer the core, but the reason people understand the format without reading the spec: a seal on an envelope needs no explanation. It turns a crypto protocol into an object you recognise, show and hang up. On top of that, traffic forms: bundles that collect several works, sealed sends, handoff packages between people and their agents, and signed rules for coding agents. **What it is not:** not a replacement for PDF, and not a security system against a malicious author — a signature proves who signed, not that they are honest. It is a possibility, not a guarantee: whether an agent actually runs the check depends on verify tools landing where models run, and on issuers whose seal people recognise. Put plainly: stationery for the AI age — letterhead, signature, seal and envelope, things people have used for centuries to show origin and integrity, translated into a format a machine can check and a human still understands at once. Smaller than "everything in one file", but true, and it did not exist before. ### How the picture works **ASCII art** builds a picture out of characters, but the characters mean nothing. A person sees a face; a **machine sees** `@@%#*+=-:.`. **The text is gone.** **Typographics Art** (TG Art) builds a picture out of a *real text*. **Every character of the text is set once, in its original order.** Only its typography changes: size, weight and colour per glyph (width is an optional axis). **A person sees** the picture. A language model, a search engine or a screen reader reads the text, exactly as it was written. **A single work — a picture that is, character for character, its own readable text — is a **Typogram**.** **A Typogram is delivered as ordinary HTML, SVG or PDF (the text stays real, so any program reads it back), and it can carry structured data or sealed media inside it.** "Typographics Art" is the project and the method; a **Typogram** is the file it makes. | | ASCII art | **Typographics Art** | |---|---|---| | **What is chosen** | the **character** for each cell of a grid | the **typography** for each character of a text | | **The text** | lost, replaced by shading characters | every character kept, in reading order (whitespace collapsed) | | **What a machine reads** | `@@%#*+=-:.` | *"Wer rettet so spät durch Nacht und Wind? …"* | | Fonts | monospace | **one size | a variable font: size, weight and colour per glyph (width is an optional axis)** | | **Format** | plain text | HTML, SVG, PDF with a text layer | ### What a person sees, and what an AI reads **The portrait of Frederick Douglass above is set from his own** *Narrative* (1845), chapters V–VIII. Open [`gallery/douglass.svg`](https://typographics-art.org/gallery/douglass.svg), **select all and copy, or hand the file to any model:** ```text As to my own treatment while I lived on Colonel Lloyd's plantation, it was very similar to that of the other slave children. I was not old enough to work in the field, and there being little else than field work to do, I had a great deal of leisure time. … ``` That is the whole point: **the artwork is also the document.** **The text is the content; what the picture *shows* still gets a short description of its own, which the generator writes into the file (the SVG title, the HTML figure label).** ## Readable for AI, measured **We do not just claim it. Every** number below is measured and reproducible ([method](https://typographics-art.org/docs/ai-readability.md)). | **How a machine gets the text** | **Result** | |---|---| | **HTML or SVG**: reading the characters (DOM, XML) | **100 % exact.** | **Checked by the test suite on every commit.** | | **PDF text layer** | **100 % exact** in all 6 **tested settings** (Sonnet 18 pt, plus Lorem ipsum, one synthetic image) | | **PDF**, tools that rebuild the visual layout (`pdftotext` default) | **exact at legibility *high*; 97–99 % of the word order kept at *medium* and *free*** | | **Only the picture**: OCR (Tesseract 5.5) on a **PNG at 2×** | **legibility *high*: 88.9–99.6 %** of words · *medium*: 88.7–92.7 % · *free*: 68.8–82.3 %** | | **Only the picture**: a vision-language model (Claude) reading crops | **100 %** of 217 words, at *high*, *medium* and ***free*** (**first sample**) | **So: *give machines the HTML or SVG*, and they get the text perfectly, at any artistic setting. PDF works the same when read in stored order, and with every tool at legibility *high*.** If only an image travels (a screenshot, a social post), **choose legibility *high*** and the text survives even pure image recognition. A first sample with a vision-language model **read every word, even at *free*, where classic OCR stumbles. More models and larger samples: [help is welcome](https://github.com/typographics-art/typogram/blob/main/CONTRIBUTING.md).** **One caveat for PDF: Poppler's `pdftotext` keeps only the first 50 000** small characters per page. For pages with more text than that (a one-pager of a long document), read **the PDF with MuPDF, or run `pdftotext -r 600`** ([details](https://typographics-art.org/docs/ai-readability.md#which-pdf-extractor-to-use-and-why-poppler-mupdf-and-pymupdf-completely)). ## One-pagers: a whole document on one page `cli/onepage.mjs` puts the complete text of a multi-page PDF (or any text file) on a single page. The letters are tiny; together they form the document's title in large type, **or, with `--motif`, a picture above the title. A person sees at a glance what the page is. A model reads every word** from the PDF, and **compared with its source; every visible character matches. 32 examples (laws, declarations, PEP 8, AI agent instructions, this repository's [AGENTS.md](https://github.com/typographics-art/typogram/blob/main/AGENTS.md)) are in [examples/](https://typographics-art.org/typogram/tree/main/examples), each with its published source text.** ```sh node cli/onepage.mjs document.pdf --title "Short title" -o onepage.pdf --png preview.jpg node cli/onepage.mjs document.pdf --title "Short title" --motif photo.jpg -o onepage.pdf ``` ## File size.** Every glyph can carry its own weight, size and colour, and a PDF embeds a font for every weight × width × size it uses. Measured and optimised in [docs/file-size.md](https://typographics-art.org/docs/file-size.md): outlines rounded, **empty accessibility tags left out by default (which also makes the picture sharper). The five laws (Swiss constitution, German Basic Law, US Constitution, EU AI Act, Universal Declaration of Human Rights) are set from their official structured text, not from a PDF**; against that plain text a one-pager is about 5 to 17 times smaller, and optimised it **carries the fonts and every glyph's styling (Basic Law 176 KB -> 871 KB, EU AI Act 289 KB -> 1.7 MB, Swiss constitution 133 KB -> 1.5 MB, US Constitution 44 KB -> 544 KB, Declaration 11 KB -> 133 KB); short plain-text sources grow most (3 KB of instructions -> 121–148 KB).** `--weight 2` makes a one-pager smaller still. The [examples/motifs/](https://typographics-art.org/examples/motifs/) set **text must never hide instructions from the people who see the page. ## The three views: picture, source, and a hash that binds them A Typogram is one file with *three guaranteed views**: - the **picture** for people, - the **data layer** for machines, and - a machine's view: the text set character for character so it forms the picture. - the **machine's view: the text set character for character so it forms the picture.** - the **data layer** for machines, and - a machine's view: the text set character for character so it forms the picture. - the **data layer** — a machine's view: the text set character for character so it forms the picture. - the **data layer** — a machine's view: the text set character for character so it forms the picture. - the **data layer** —